

# International Rocket Engineering Competition

## *2027 Rules & Requirements*

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*This electronic version is an official and approved document.*
*Verify that this is the latest revision before use.*

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# 1.0 PURPOSE, USE, AND DOCUMENT CONVENTIONS

## 1.1 PURPOSE

This specification reorganizes the 2027 International Rocket Engineering Competition (IREC) Rules & Requirements into numbered, subject-led paragraphs. It is intended to help student teams understand what is required, prepare their technical reports and presentations, and make the same content easier for judges to evaluate.

The conversion preserves the substantive intent of the source document while separating requirements, permissions, recommendations, scoring provisions, and explanatory guidance. A small number of derived requirements are included where needed to make the Project Technical Report content consistent with the judging rubrics.

## 1.2 APPLICABILITY

R&R-1.2.1 **Specification Compliance**

Each participating team shall comply with this specification, the applicable documents identified in Section 2.0, and current instructions issued by ESRA for the competition year.

R&R-1.2.2 **Current Document Versions**

Each team shall verify that it is using the current, officially approved electronic version of all IREC-controlled documents before use.

R&R-1.2.3 **Requirement Clarification**

A team requiring clarification of a requirement, its spirit or intent, or a situation not addressed by the specification shall request clarification through the ESRA communication channel identified for the competition year.

## 1.3 CONVENTION AND NOTATION

The following definitions differentiate between requirements and other statements. The degree to which a team satisfies the spirit and intent of these statements will guide the competition officials' decisions on a project's flight status and overall score at the IREC.

**Shall** This is the only verb used to denote mandatory requirements. Failure to satisfy the spirit and intent of a mandatory requirement will always affect a project's score and flight status.

**May** This verb signifies that teams are allowed, but not required, to exercise the associated action.

**Should** This verb is used for stating non-mandatory goals. Failure to satisfy the spirit and intent of a non-mandatory goal may affect a project's score and flight status, depending on design implementation and the team's ability to provide thorough documentary evidence of their due diligence on-demand.

**Will** This verb is used for stating facts and declarations of purpose. The authors use these statements to clarify the spirit and intent of requirements and goals.

## 1.4 REVISIONS

The IREC R&R may require revision periodically (from time to time or from one competition to the next), based on the lessons learned by both host organizations and the participants.

- Major revisions will be accomplished by complete document reissue.
- "Real world events" may require smaller revisions to this document in the months leading up to a competition.
- Revisions will be reflected in updates to the document's effective date.

The authority to approve and issue revised versions of this document rests with ESRA.

# 2.0 APPLICABLE AND REFERENCE DOCUMENTS

The documents in Table 2-1 are applicable to the extent invoked by this specification or provide supporting guidance and forms. Links are retained from the source baseline because ESRA may update the linked documents from year to year.

**Table 2-1 -** Applicable and Reference Documents

| DOCUMENT | FILE LOCATION | APPLICABILITY |
|---|---|---|
| IREC Design, Test, & Evaluation Guide (DTEG) | https://www.esrarocket.org/documents-and-forms | Safety, design, test, and evaluation requirements and guidance invoked throughout this specification. |
| IREC Integrated Master Schedule (IMS) | https://www.esrarocket.org/documents-and-forms | Competition deliverable milestones and deadlines. |
| IREC Range Standard Operating Procedures | https://www.esrarocket.org/documents-and-forms | Range conduct and operational requirements. |
| IREC Team Video Challenge | https://www.esrarocket.org/team-video-challenge | Video challenge requirements. |
| IREC Live Rocket Video Challenge | https://www.esrarocket.org/video-challenge | Live-video challenge requirements. |
| IREC Project Technical Report Template | https://www.esrarocket.org/documents-and-forms | Required Technical Report formatting baseline. |
| IREC Extended Abstract Template | https://www.esrarocket.org/documents-and-forms | Required Extended Abstract formatting baseline. |
| TRA Unified Safety Code | https://www.tripoli.org/docs/Safety_Code | Applicable safety code as invoked by IREC guidance. |
| 14 CFR Part 1, §1.1 General Definitions | http://www.ecfr.gov/General_Definitions | Federal definitions. |
| 14 CFR Part 101, Subpart C, §101.22 Definitions | http://www.ecfr.gov/Part101_SubpartC | Amateur rocket definitions. |

# 3.0 COMPETITION MISSION AND SYSTEM REQUIREMENTS

## 3.1 COMPETITION MISSION AND CATEGORIES

R&R-3.1.1 **Competition Mission**

Each competing team shall design, build, and launch a rocket that carries a payload of not less than 4.4 *lbs* to the target apogee associated with its approved competition category.

R&R-3.1.2 **Competition Category**

Each team project shall be assigned to one of the categories listed in Table 3-1.

**Table 3-1 -** IREC Project Categories

| CATEGORY | TEAM ID | TARGET APOGEE | PROPULSION / CONFIGURATION | SCORED |
|---|---|---|---|---|
| COTS 10K | 1-199 | 10,000 ft AGL | COTS Solid or Hybrid | Yes |
| COTS 30K | 200-299 | 30,000 ft AGL | COTS Solid or Hybrid | Yes |
| SRAD Solid 10K | 300-399 | 10,000 ft AGL | SRAD Solid | Yes |
| SRAD Solid 30K | 400-499 | 30,000 ft AGL | SRAD Solid | Yes |
| Two-Stage 30K | 500-599 | 30,000 ft AGL | Two-Stage; any allowed propulsion type | Yes |
| Two-Stage 45K | 600-699 | 45,000 ft AGL | Two-Stage; any allowed propulsion type | Yes |
| SRAD Hybrid 10K | 700-749 | 10,000 ft AGL | SRAD Hybrid | Yes |
| SRAD Hybrid 30K | 750-799 | 30,000 ft AGL | SRAD Hybrid | Yes |
| SRAD Liquid 5K | 800-849 | 5,000 ft AGL | SRAD Liquid | Yes |
| SRAD Liquid 10K | 850-899 | 10,000 ft AGL | SRAD Liquid | Yes |
| Demonstration Flight | 900-999 | As approved | Non-competing, including approved high-altitude demonstrations of 50,000 ft or more | No |

R&R-3.1.3 **Category Downgrade**

A team requesting a category change shall request only a downgrade from its accepted category; category upgrades are not permitted.

R&R-3.1.4 **Category Change Approval**

A team shall obtain ESRA approval for a category change before submitting its final Project Technical Report.

R&R-3.1.5 **Category Change Request**

A category-change request shall be submitted to general.info@esrarocket.org, comprules@esrarocket.org and launchops@esrarocket.org with an email subject that identifies the team ID, the formal category-change action, and the competition year.

R&R-3.1.6 **Post-Report Category Change**

After submission of the final Project Technical Report, a team shall not change categories except to an ESRA-approved non-competing demonstration-flight category.

## 3.2 General Design, Propulsion, and Flight-Safety Requirements

R&R-3.2.1 **Student-Developed Propulsion**

An SRAD propulsion system shall be designed and manufactured by students on the team.

R&R-3.2.2 **Student Propellant Manufacture**

SRAD propellant components and energetic materials, including solid propellant grains, shall not be manufactured by a third party.

R&R-3.2.3 **Non-Toxic Propellants**

All propellants used by an IREC project shall be non-toxic as defined in Appendix A.

R&R-3.2.4 **Safety and Regulatory Compliance**

Each project shall comply with applicable requirements of the DTEG, the IREC Range Standard Operating Procedures, NFPA 1127, FAA regulations, and other regulatory requirements invoked by ESRA.

NOTE: Multi-stage launch vehicles and solid, liquid, and hybrid chemical propulsion disciplines are permitted, subject to the approved category and applicable safety requirements. Departures from DTEG guidance may affect score or flight status in proportion to severity.

## 3.3 Team Composition and Eligibility

R&R-3.3.1 **Student Status**

Team members shall have been matriculated undergraduate or graduate students during the academic year immediately preceding the competition.

R&R-3.3.2 **Single-Team Participation**

An individual student shall participate on no more than one IREC team accepted during the competition year.

R&R-3.3.3 **Independent Teams**

Each accepted team shall operate as an independent project team and shall not share student team members with another accepted team.

R&R-3.3.4 **Single Competition Category**

A project shall be entered in no more than one IREC category.

R&R-3.3.5 **Two Teams from One Institution**

An academic institution entering two teams shall use different propulsion forms for the two teams, except that a non-competing demonstration or approved high-altitude demonstration team may use the same propulsion form as the institution's competing team.

R&R-3.3.6 **Multi-School Team Approval**

A proposed multi-school team shall obtain ESRA approval before the team is formed and accepted for competition.

R&R-3.3.7 **Multi-School Participation Restriction**

When an institution fields its own IREC team, students from that institution shall not join a different multi-school team.

R&R-3.3.8 **Institutional Affiliation Documentation**

A multi-school team shall document each institutional affiliation through the School Participation Letters required by Section 4.8.1.

NOTE: There is no limit on total team size or on the number of graduate students. Teams are strongly encouraged to obtain relevant high-power rocketry experience, use an experienced mentor and Flyer of Record, and demonstrate project-relevant experience such as full-scale static firing for SRAD, hybrid, or liquid systems. These factors may be considered during team acceptance.

## 3.4 PAYLOAD REQUIREMENTS

### 3.4.1 PAYLOAD MASS AND DEFINITION

R&R-3.4.1.1 **Minimum Payload Mass**

The launch vehicle shall carry a payload with a nominal mass of not less than 4.4 *lbs*.

R&R-3.4.1.2 **Payload Definition**

The payload shall be replaceable by ballast of the same mass without changing the launch vehicle trajectory to target apogee or the success of launch-vehicle recovery.

R&R-3.4.1.3 **Payload and Stability Analysis**

The launch vehicle stability analysis may include the required payload mass; stability without the required payload installed is not required.

R&R-3.4.1.4 **Official Payload Weigh-In**

Official payload mass shall be determined using the competition-official scale.

R&R-3.4.1.5 **Official Mass Tolerance**

A payload measured by the official scale at not less than 95 percent of the 4.4 *lbs* minimum shall be accepted without a mass penalty.

R&R-3.4.1.6 **Payload Structural Integrity**

The payload shall remain mechanically secure and undamaged throughout launch, flight, and recovery.

### 3.4.2 INDEPENDENT PAYLOAD FUNCTIONALITY

R&R-3.4.2.1 **Independent Vehicle Performance**

The launch vehicle shall achieve target apogee and recover successfully without relying on any active or passive payload function.

R&R-3.4.2.2 **Launch-Vehicle Functions**

A function required for launch-vehicle stability, control, recovery, or mission success shall be classified as a launch-vehicle subsystem rather than as payload.

NOTE: Non-functional ballast payloads are permitted. Functional scientific experiments and technology demonstrations are encouraged and may be entered in the SDL Payload Challenge under separately published rules.

### 3.4.3 PAYLOAD LOCATION, INTERFACE, AND CHECK-IN

R&R-3.4.3.1 **Two-Stage Payload Location**

For a two-stage category, the payload shall be located in the sustainer.

R&R-3.4.3.2 **Independent Payload Weighing**

The payload shall be removable and weighable independently of launch-vehicle recovery systems, internal structure, airframe, and other launch-vehicle components.

R&R-3.4.3.3 **Non-Removable Payload Interface**

Any payload adapter or interface that cannot be separated for official weighing shall be treated in accordance with the payload scoring provisions and may result in a point penalty.

R&R-3.4.3.4 **Payload Load Capability**

The payload and its retention system shall withstand applicable launch and recovery loads without shifting.

R&R-3.4.3.5 **SDL Payload Check-In**

Each payload, including non-functional payloads and payloads not entered in the SDL Payload Challenge, shall be weighed, measured, and checked in by SDL before launch.

### 3.4.4 RESTRICTED PAYLOAD MATERIALS

R&R-3.4.4.1 **Heavy Metals**

A payload shall not contain significant quantities of lead or other heavy metals.

R&R-3.4.4.2 **Hazardous Payload Materials**

A payload shall not contain hazardous material that creates an unacceptable health or safety impact to team members, staff, the public, the conference venue, or the launch site.

R&R-3.4.4.3 **Radioactive Material Approval**

Radioactive material shall be used only when operationally necessary and approved by competition officials before the event.

R&R-3.4.4.4 **Radioactive Material Limit**

Approved radioactive material shall be fully encapsulated and shall have activity not greater than 1 microcurie.

R&R-3.4.4.5 **Live Animals**
A payload shall not contain a live vertebrate animal.

### 3.4.5 PAYLOAD FORM FACTOR

**Table 3-2 -** Payload Cube Unit Form-Factor Definitions

| PAYLOAD CUBE UNITS | DIMENSIONS (L × W × H) | LENGTH/WIDTH TOLERANCE | HEIGHT TOLERANCE |
|---|---|---|---|
| 1U | 100 mm × 100 mm × 100 mm | ± 2 mm | ± 2 mm |
| 2U | 100 mm × 100 mm × 200 mm | ± 2 mm | ± 2 mm |
| 3U | 100 mm × 100 mm × 300 mm | ± 2 mm | ± 2 mm |
| 4U | 100 mm × 100 mm × 400 mm | ± 2 mm | ± 2 mm |
| 5U | 100 mm × 100 mm × 500 mm | ± 2 mm | ± 2 mm |
| 6U | 100 mm × 100 mm × 600 mm | ± 2 mm | ± 2 mm |

NOTE: To comply with the Payload Cube Unit dimensions, the payload outer mold line must be completely inscribed by a rectangular prism at the minimum tolerance condition and circumscribed by a rectangular prism at the maximum tolerance condition. A 100 mm diameter rod is therefore not compliant, while a 100 mm cube is compliant

R&R-3.4.5.1 **Non-Functional Payload Form**

A launch vehicle carrying only non-functional payload mass shall carry that payload as a 3U Payload Cube Unit conforming to Table 3-2.

R&R-3.4.5.2 **Functional Payload Form**

A functional scientific experiment or technology-demonstration payload may use any form factor provided it complies with the mass, independence, interface, and restricted-material requirements of Sections 3.4.

R&R-3.4.5.3 **Functional Payload Mass:**

A functional payload shall achieve the required payload mass primarily through the experiment, technology, and associated support structure.

R&R-3.4.5.4 **Boilerplate Mass Limit**

A functional payload may include not more than 1.15 *lbs* of non-functional boilerplate mass to meet the minimum payload mass.

R&R-3.4.5.5 **Boilerplate Mass Weighing**

Non-functional boilerplate mass used with a functional payload shall be separately removable and separately weighed.

R&R-3.4.5.6 **Payload Cube Unit Bonus**

A functional payload claimed for the Payload Cube Unit bonus shall fit completely within the applicable Payload Cube Unit dispenser envelope with no protrusion or physical connection outside the envelope.

### 3.4.6 DEPLOYABLE PAYLOADS

R&R-3.4.6.1 **Deployable Payload Tracking**

A deployable payload that separates during flight or after landing shall include GPS tracking as required by the DTEG.

R&R-3.4.6.1 **Prohibited UAS Payloads**

A deployable payload shall not be an unmanned aircraft system (UAS).

NOTE: Deployable payloads introduce significant project risk and will receive heightened safety scrutiny.

## 3.5 DEMONSTRATION FLIGHTS REQUIREMENTS

R&R-3.5.1 **High-Altitude Airspace Coordination**

A team planning a high-altitude demonstration flight that may require a Class 3 waiver shall coordinate with ESRA officials to secure required airspace permissions.

R&R-3.5.2 **Demonstration-Flight Deliverables**

A non-competing demonstration-flight team shall submit progress updates, a Project Technical Report, a School Participation Letter, and proof of insurance.

NOTE: Non-competing demonstration-flight teams are not eligible for category awards, technical awards, or podium sessions.

## 3.6 TRACKING AND OFFICIAL ALTITUDE LOGGING REQUIREMENTS

R&R-3.6.1 **Rocket GPS Tracking**

Each IREC launch vehicle shall carry a GPS tracking system that complies with the DTEG.

R&R-3.6.2 **Official Altitude Logger**

Each launch vehicle shall carry a COTS barometric-pressure altimeter with onboard data storage for the official apogee log.

R&R-3.6.3 **Altitude Logger Configuration**

The official altitude logger may be a standalone COTS device or a COTS flight computer also used for recovery-system deployment.

R&R-3.6.4 **Telemetry Sensor Source**

When telemetry is used as a secondary source of official altitude, the telemetry data shall originate from the same sensor source as the official onboard data log.

R&R-3.6.5 **Postflight Reporting**

Immediately after recovery, the recovery team shall report directly to Postflight Inspection with the rocket and all tools, cables, computers, and other equipment needed to access the official altimeter data without relying on site power or internet access.

R&R-3.6.6 **Official Altitude Source**

The first COTS altimeter selected by the team for audible or electronic readout shall be the official altitude source unless that altimeter is damaged and its data is unreadable.

R&R-3.6.7 **Postflight Data Access**

The team shall permit competition officials to open or disassemble the electronics bay as needed to verify flight data.

R&R-3.6.8 **Postflight Inspection Time Limit**

The team shall complete the Postflight Data Inspection within 45 minutes; exceeding this period due to unpreparedness shall incur a 20-point penalty.

R&R-3.6.9 **Postflight Check-In Deadline**

A team shall check in with its recovered rocket for Postflight Data Inspection by 20:00 local time Wednesday through Friday and by 09:00 local time Saturday.

R&R-3.6.10 **Saturday Scoring Cutoff**

A team that has not received a recovery score by 10:00 local time Saturday shall receive a zero flight-altitude score.

# 4.0 PROJECT DELIVERABLE REQUIREMENTS

## 4.1 GENERAL SUBMISSION REQUIREMENTS

R&R-4.1.1 **Required Deliverable Submission**

Each team shall submit all required deliverables through the locations and methods specified by ESRA.

R&R-4.1.2 **Deliverable Content and Format**

Each deliverable shall comply with its specified format, content, naming, and submission requirements.

R&R-4.1.3 **Deliverable Deadlines**

Each required deliverable shall be submitted by the deadline in the current IMS.

R&R-4.1.4 **Late Submission Disqualification**

A late required submission shall result in team disqualification unless ESRA issues a written exception.

R&R-4.1.5 **Incomplete or Incorrect Submissions**

Incomplete or incorrect submissions, including data entered in the wrong fields, incorrect units, or violations of submission guidance, shall be subject to the penalties specified in Section 6.7.

## 4.2 ENTRY FORM AND PROGRESS UPDATES

R&R-4.2.1 **Prospective Team**

A prospective team shall register as a new team through esrarocket.org to communicate its intent to compete.

R&R-4.2.2 **Three Design Milestone Submissions**

Each team shall submit and pass three design milestone gates through esrarocket.org on the occasions identified in the IMS.

R&R-4.2.3 **Best Available Information**

The Entry Form and each Design Milestone shall be complete, timely, and populated with the best available technical information.

R&R-4.2.4 **Uncertain or Preliminary Data**

When a value is uncertain, estimated, unreasonable, or not yet decided, the team shall still provide information and shall identify the estimate, uncertainty, ongoing analysis, or applicable guidance in the available comment fields.

R&R-4.2.5 **Team ID in Correspondence**

After a Team ID is assigned, team correspondence with ESRA shall include the Team ID.

R&R-4.2.6 **File Naming Convention**

Progress-update files shall use the naming convention TeamID_Formal University Name_Purpose.

R&R-4.2.7 **Filename Penalty**

A progress-update submission using a nonconforming filename shall incur a 5-point deduction from that submission.

## 4.3 ONLINE VIDEO SAFETY REVIEW

R&R-4.3.1 **Online Video Safety Review**
For the third Progress Update, the team shall participate in an online safety review with its assigned safety reviewers.

R&R-4.3.2 **Safety Review Presentation**
The team shall prepare a short slide presentation that reviews current progress and addresses issues previously identified by safety reviewers.

R&R-4.3.3 **Rocket Availability**
The team shall make the rocket available for visual review during the third Progress Update safety review.

R&R-4.3.4 **Safety Review Coordination**
The team shall coordinate and provide the online meeting link and shall invite its Flight Safety Reviewer or reviewers, Flyer of Record, and Mentor.

R&R-4.3.5 **Required Safety Review Attendees**
The Flight Safety Reviewer or reviewers, Flyer of Record, and Mentor shall be present at the online safety review.

R&R-4.3.6 **Safety Review Recording**
The team shall record the online safety review and provide the recording to ESRA for publication on ESRA social-media platforms.

R&R-4.3.7 **Safety Issue Resolution**
The team shall satisfactorily resolve safety, DTEG, and rules-compliance issues raised during progress-update reviews.

NOTE: A team with significant production work remaining at the third Progress Update, or a team unable to resolve safety concerns, may be disqualified. Online safety reviews do not replace hands-on event safety review and RSO processes.

## 4.4 PROJECT TECHNICAL REPORT

### 4.4.7 GENERAL FORMAT AND SUBMISSION

R&R-4.4.1 **Project Technical Report**

Each team shall submit a Project Technical Report that describes the project for the judging panel and competition officials.

R&R-4.4.2 **Report Template and Style**

The Project Technical Report shall use the current ESRA template and the AIAA style guide identified in Section 2.0.

R&R-4.4.3 **Equivalent Report Formatting**

A report prepared using software other than Microsoft Word shall reproduce the formatting of the current ESRA template.

R&R-4.4.4 **COTS Report Page Limit**

The main body of a COTS-team Project Technical Report shall not exceed 25 pages, including tables and graphics, this page limit starts at the introduction and ends at the conclusion.

R&R-4.4.5 **SRAD Report Page Limit**

The main body of an SRAD solid, hybrid, or liquid-team Project Technical Report shall not exceed 35 pages, including tables and graphics, this page limit starts at the introduction and ends at the conclusion.

R&R-4.4.6 **Appendix Length**

Project Technical Report appendices may be of any length.

R&R-4.4.7 **Report Submission**

The team shall submit one digital PDF copy of the Project Technical Report by the IMS deadline through esrarocket.org.

R&R-4.4.8 **Large Report Access**

When a Project Technical Report exceeds 50 MB and is hosted externally, the team shall provide judges unrestricted access to the file.

R&R-4.4.9 **Required Subtitle**

The Project Technical Report shall use the subtitle “Team <Team ID> Project Technical Report to the <Year> IREC.”

R&R-4.4.10 **Public Release**

The team shall not mark the Project Technical Report or Extended Abstract as confidential or “Do Not Share.”

NOTE: Teams should bring a limited number of hard copies to the event for consultation by judges and competition officials. All submitted reports will be publicly published after the event.

### 4.4.8 AUTHORSHIP AND CONTACT INFORMATION

R&R-4.4.2.1 **Report Authors**

The Project Technical Report shall list each author who made a significant contribution and shall include the team’s academic advisor.

R&R-4.4.2.2 **Author Affiliations**

The report shall state each listed individual’s academic affiliation using an address format appropriate to the country and including institution, city, state or region, postal code, and country where applicable.

R&R-4.4.2.3 **Corresponding Author**

The report shall identify a corresponding author and provide an email address at which that author can be contacted for at least two years after submission.

### 4.4.9 EXECUTIVE SUMMARY

R&R-4.4.3.1 **Executive Summary Length**

The Project Technical Report shall contain an Executive Summary not exceeding one page.

R&R-4.4.3.2 **Executive Summary Content**

The Executive Summary shall identify the competition category, major rocket components, payload mission, unique design features, required testing, and testing performed.

### 4.4.10 INTRODUCTION AND PROJECT DEFINITION

R&R-4.4.4.1 **Introduction Length**

The Project Technical Report shall contain an Introduction not exceeding two pages.

R&R-4.4.4.2 **Introduction Content**

The Introduction shall identify the competition category, unique launch-vehicle design characteristics, payload mission, and high-level project or program goals and objectives.

R&R-4.4.4.3 **Academic Program and Contributors**

The Introduction shall describe the academic program, external contributors and facilities, and funding sources associated with the project.

R&R-4.4.4.4 **Team Organization Chart**

The Introduction shall include a team organization chart and summarize team-management strategies.

R&R-4.4.4.5 **Report Preparation Process**

The Introduction shall describe how the Project Technical Report was authored, integrated, reviewed, and edited.

### 4.4.11 SYSTEM ARCHITECTURE, INTERFACES, AND DESIGN RATIONALE

R&R-4.4.5.1 **System Architecture Overview**

The Project Technical Report shall contain a System Architecture Overview beginning with a top-level description of the integrated launch vehicle.

R&R-4.4.5.2 **System Architecture Cutaway**

The System Architecture Overview shall include a cutaway figure of the fully integrated, mission-configured launch vehicle and shall identify its major subsystems.

R&R-4.4.5.3 **Subsystem Coverage**

The report shall describe the aero-structures, propulsion, avionics, recovery, telemetry, payload, and other flight-critical subsystems, or explicitly state when a listed subsystem is not applicable.

R&R-4.4.5.4 **Subsystem Description**

Each subsystem description shall present the engineering analyses and rationale supporting design and manufacturing decisions, including applicable goals, constraints, alternatives, trade studies, anticipated loads, and safety factors.

R&R-4.4.5.5 **System Interfaces**

The report shall identify and describe the mechanical, electrical, software, fluid, functional, and operational interfaces that are significant to integrated mission success.

R&R-4.4.5.6 **Assumptions, Margins, and Sensitivity**

For each major performance analysis, the report shall identify material assumptions, input uncertainties, design margins, and the sensitivity of conclusions to the variables that most strongly affect the result.

R&R-4.4.5.7 **Model Basis and Limitations**

The report shall distinguish analysis predictions from test-validated results and shall identify the basis, limitations, and applicability of each analytical model.

R&R-4.4.5.8 **Engineering Drawings**

Engineering and technical drawings supporting subsystem definitions shall be included in Appendix F.

### 4.4.12 MISSION CONCEPT OF OPERATIONS

R&R-4.4.6.1 **Mission CONOPS**

The Project Technical Report shall contain a Mission Concept of Operations overview.

R&R-4.4.6.2 **Mission Phases**

The CONOPS shall identify mission phases, include a mission-phase figure, and describe the nominal operation and responsible subsystem or subsystems during each phase.

R&R-4.4.6.3 **CONOPS**

The CONOPS shall define the events or conditions that initiate and terminate each mission phase.

R&R-4.4.6.4 **Mission Success Criteria**

The mission phases defined by the CONOPS shall be used to organize mission-success failure modes in the Risk Assessment.

### 4.4.13 VERIFICATION, VALIDATION, AND EVIDENCE

R&R-4.4.7.1 **Integrated Evidence Summary**

The Project Technical Report shall summarize the analyses, inspections, tests, and demonstrations used to establish that the integrated design is ready for its intended mission.

R&R-4.4.7.2 **For Major Test or Demonstration Report**

For each major test or demonstration, the report shall explain what design question, risk, model, or performance objective the activity was intended to address.

R&R-4.4.7.3 **Test Configuration, Acceptance Criteria, Results, Anomalies and Conclusions**

The report shall identify test configuration, acceptance criteria, results, anomalies, conclusions, and resulting design actions for each reported verification or validation activity.

### 4.4.14 CONCLUSIONS AND LESSONS LEARNED

R&R-4.4.8.1 **Conclusions and Lessons Learned**

The Project Technical Report shall contain a Conclusions and Lessons Learned section.

R&R-4.4.8.2 **Technical and Team Management Lessons Learned**

The Conclusions and Lessons Learned shall address technical and team-management lessons from design, manufacturing, testing, and failures.

R&R-4.4.8.3 **Transfer of Knowledge**

The Conclusions and Lessons Learned shall describe strategies for transferring corporate knowledge from senior students to future team members.

### 4.4.15 ACKNOWLEDGMENTS

R&R-4.4.9.1 **Acknowledgments**

The Project Technical Report shall acknowledge key individuals, factuality advisors, institutions, facilities, and funding sources.

### 4.4.16 TECHNICAL REPORT APPENDICES

R&R-4.4.10.1 **Required Appendices:**

The Project Technical Report shall include Appendices A through F in the order specified in Sections 4.4.16.

## Appendix A — System Weights, Measures, and Performance Data

R&R-4.4.10.2 **Consistent Units**

Appendix A shall use either an Imperial or Metric unit system consistently; when manufacturer data use a different system, the report shall include a conversion to the report's selected unit system.

**Table 4-1 -** Minimum Appendix A Data Set

| DATA GROUP | REQUIRED DATA |
|---|---|
| Basic vehicle information | Number of stages; length; airframe diameter; number of fins; fin semi-span; fin tip chord; fin root chord; fin thickness; weight on launch rail; propellant weight; empty motor case/structure weight; payload weight; liftoff weight; center of pressure; center of gravity; launch rail length. |
| Propulsion information | Motor type; COTS or SRAD classification; COTS manufacturer and designation; motor letter classification; average thrust; total impulse; motor burn time. |
| Predicted flight data | Liftoff thrust-to-weight ratio; rail-departure velocity; minimum static margin; maximum acceleration; maximum velocity; fin-flutter velocity; target apogee; predicted apogee. |
| Flight profile | A flight-profile graph with clearly identified variables, units, key events, and predicted values needed to understand the mission. |
| Recovery information | COTS and redundant altimeters; drogue and main parachute primary/backup deployment charges, deployment altitudes, and descent rates; parachute shock cords; mechanical links. |

R&R-4.4.10.3 **Required System Data**

Appendix A shall contain all applicable data listed in Table 4-1 and shall mark non-applicable items as not applicable rather than omitting them without explanation.

### Appendix B — Project Test Report

R&R-4.4.10.4 **Applicable Test Reports Required**

Appendix B shall contain the applicable test reports required by the current DTEG in the prescribed order.

R&R-4.4.10.5 **Not-Applicable Test Reports**

When a prescribed test report is not applicable, the team shall include a page marked “THIS PAGE INTENTIONALLY LEFT BLANK” in its place.

R&R-4.4.10.6 **Recovery Electronics Redundancy**

Recovery-system test reporting shall address DTEG recovery-system, ground-test, flight-test, stored-energy-device, arming-device, and arming-verification requirements and shall include a figure and explanation of dual redundancy in recovery electronics.

R&R-4.4.10.7 **Fluid Circuit Diagram**

An SRAD hybrid or liquid propulsion-system test report shall include a fluid-circuit diagram identifying nominal operating pressure at key points, including the fill system.

R&R-4.4.10.8 **Additional Applicable Test Reports**

Appendix B shall include applicable SRAD pressure-vessel, SRAD GPS, and payload-recovery-system test reports.

## Appendix C — Hazard Analyses

R&R-4.4.10.9 **Hazard Analysis**

Appendix C shall contain a hazard analysis addressing applicable hazardous-material handling, transportation, storage, propellants, and other design or operational hazards to personnel.

R&R-4.4.10.10 **Design Mitigation Strategies**

The hazard analysis shall define a process or design mitigation for each identified hazard.

## Appendix D — Risk Assessment

R&R-4.4.10.11 **Risk Assessment**

Appendix D shall contain a risk assessment summarizing project risk and reliability concepts.

R&R-4.4.10.12 **Failure Mode Records**

The risk assessment shall record failure modes that threaten mission success in a matrix organized by CONOPS mission phase.

R&R-4.4.10.13 **Design Mitigation Strategies**

The risk assessment shall define a process or design mitigation for each identified risk.

## Appendix E — Checklists

R&R-4.4.10.14 **Checklists**

Appendix E shall contain detailed, step-by-step checklists for final assembly, preflight, arming, launch, recovery, off-nominal operations, and any additional procedures required by special design features.

R&R-4.4.10.15 **Range Conduct**

Off-nominal checklists shall include alternate disarming and safeing process flows based on identified failure modes and shall not conflict with the IREC Range Standard Operating Procedures.

R&R-4.4.10.16 **Fill and Vent Procedures**

An SRAD hybrid or liquid team shall include fill, vent, fault-response, tank-cleaning, and fluid-circuit-component-cleaning procedures.

R&R-4.4.10.17 **Printed Checklists**

The team shall maintain a complete paper hardcopy of applicable checklist procedures with the flight hardware during all range activities.

R&R-4.4.10.18 **Detailed Checklists**

The team shall use its detailed checklists during all operations for which checklists are provided.

NOTE: Insufficient checklist detail, failure to bring paper hardcopies, or failure to use the checklists will make the team ineligible for flight activity until the deficiency is resolved.

### Appendix F — Engineering Drawings

R&R-4.4.10.19 **Engineering Drawings**

Appendix F shall contain revision-controlled engineering drawings needed to define significant subsystems and components.

R&R-4.4.10.20 **SRAD Drawings**

Appendix F shall provide drawings for all SRAD subsystems and components.

R&R-4.4.10.21 **Drawing Content**

Each engineering drawing shall include the dimensions and specifications required to define the item.

R&R-4.4.10.22 **CAD File Size**

The Project Technical Report shall use images or drawing views rather than embedded full CAD models to control file size.

## 4.5 Poster Session Materials

R&R-4.5.1 **Poster Display**
Each team shall bring a poster display to the IREC Conference that communicates the overall project design, testing, CONOPS, and purpose.

R&R-4.5.2 **Poster Dimensions**
The poster shall be approximately 36 in × 48 in and shall be self-supporting on an organizer-provided table or a team-provided easel.

R&R-4.5.3 **Poster Team ID**
The poster shall display the Team ID in the upper-right corner in bold black Arial or similar type, size 72 point or larger, on a white field.

R&R-4.5.4 **Poster Session Participation**
Each team shall participate in the Poster Session and shall keep one or more team members with the display to answer questions and present the work.

R&R-4.5.5 **Team**
A team shall make practicable non-energetic project hardware available for display and Design/Build/Quality judging.

R&R-4.5.6 **Team Absent from Conference Day**
A team absent from Conference Day shall incur a 20-point conduct penalty.

R&R-4.5.7 **Poster Submission**
Each team shall submit a digital PDF copy of its poster by the IMS deadline using the method specified by ESRA.

R&R-4.5.8 **Poster Submission Penalty**
Failure to submit poster materials or comply with poster submission requirements shall incur a 20-point administrative penalty.

R&R-4.5.9 **Conference Energetics Prohibition**

Energetic items, including initiators, fuel grains, ejection charges, and pressurized gas cylinders, shall not be brought into the conference premises.

## 4.6 PODIUM SESSION MATERIALS

### 4.6.1 EXTENDED ABSTRACT

R&R-4.6.1.1 **Extended Abstract Topic**

Each team shall submit an Extended Abstract proposing a launch-vehicle-related technical topic for possible Podium Session selection.

R&R-4.6.1.2 **Extended Abstract Format**

The Extended Abstract shall use the current ESRA Microsoft Word template and AIAA style guide.

R&R-4.6.1.3 **Required Subtitle**

The Extended Abstract shall use the subtitle “Team <Team ID> Technical Presentation to the <Year> IREC.”

R&R-4.6.1.4 **Extended Abstract Length**

The Extended Abstract shall contain at least 500 words and shall not exceed two pages, excluding footnotes and source notes.

R&R-4.6.1.5 **Extended Abstract Content Restrictions**

The Extended Abstract shall not contain tables, figures, nomenclature lists, equations, or appendices.

R&R-4.6.1.6 **Extended Abstract Technical Content**

The Extended Abstract shall describe the topic purpose, technical foundation, preliminary results, and expected results of IREC flight testing.

R&R-4.6.1.7 **Launch-Vehicle Topic**

The Podium Session topic shall concern launch-vehicle development and shall not be limited to a payload topic.

R&R-4.6.1.8 **Extended Abstract Submission**

The team shall submit a digital PDF copy of the Extended Abstract through the ESRA-designated location by the IMS deadline.

R&R-4.6.1.9 **Extended Abstract Penalty**

Failure to submit the Extended Abstract or comply with its submission requirements shall incur a 20-point administrative penalty.

NOTE: Strong topics generally demonstrate technical excellence, innovation, rigorous modeling and simulation, verification and validation, manufacturing capability, or a well-documented solution to a unique technical challenge. The examples in the source baseline are retained in spirit but are not limiting.

### 4.6.2 PRESENTATION SLIDES AND PODIUM PARTICIPATION

R&R-4.6.2.1 **Advance Slide Submission**

At the Extended Abstract deadline, the team shall submit a digital PDF copy of the presentation slides it intends to use if selected.

R&R-4.6.2.2 **Slide Team ID**

Presentation slides shall identify the Team ID.

R&R-4.6.2.3 **Presentation Slide Penalty**

Failure to submit presentation slides or comply with slide submission requirements shall incur a 20-point administrative penalty.

R&R-4.6.2.4 **Podium Presentation Time**

A team selected for the Podium Session shall be prepared to present for 20 minutes and answer questions for an additional five minutes.

R&R-4.6.2.5 **Presentation Laptop**

A selected team shall bring a laptop computer containing the completed presentation slides.

R&R-4.6.2.6 **Missed Presentation**

A selected team that misses its assigned presentation time shall forfeit the opportunity to present.

R&R-4.6.2.7 **Professional Conduct**

A team shall not lobby for an exception after missing its Podium Session time; such conduct shall be subject to the conduct penalty provisions.

NOTE: Competition officials will select not fewer than 24 teams for the Podium Session and may use runner-up teams to replace selected teams that do not attend.

## 4.7 INTEGRATED MASTER SCHEDULE MILESTONES

R&R-4.7.1 **IMS Milestones**

Each team shall meet all applicable milestones and deadlines in the current IMS.

R&R-4.7.2 **Standard IMS Deadline Time**

Unless otherwise specified, an IMS deadline shall be 17:00 Central Time in Midland, Texas, on the date listed in the IMS.

NOTE: Failure to meet the required timeline may cause a point penalty, disqualification, or loss of flight eligibility, depending on the affected requirement.

## 4.8 ADMINISTRATIVE DOCUMENTS

### 4.8.1 SCHOOL PARTICIPATION LETTER

R&R-4.8.3.1 **Institutional Participation Letter**

Each represented academic institution shall provide a signed School Participation Letter acknowledging the team's IREC participation.

R&R-4.8.3.2 **Authorized Letter Signature**

The School Participation Letter shall be signed by a faculty member or other paid, non-student institutional representative.

R&R-4.8.3.3 **Letterhead and Attendee Roster**

The School Participation Letter shall use official institutional letterhead and shall list the students attending from the team.

R&R-4.8.3.4 **Multi-School Participation Letters**

A multi-school team shall submit one School Participation Letter from each represented institution.

R&R-4.8.3.5 **Participation Letter Submission**

The team shall submit signed School Participation Letter PDF files through the ESRA-designated location by the IMS deadline.

R&R-4.8.3.6 **Participation Letter Penalty**

Failure to submit a School Participation Letter or comply with the template or submission requirements shall incur a 20-point administrative penalty and may require resubmission.

### 4.8.2 INSURANCE

R&R-4.8.4.1 **Tripoli Membership**

A solid or hybrid team using Tripoli Launch Insurance shall ensure each attending team member is a dues-paying Tripoli Level 0 member and shall provide proof at registration.

R&R-4.8.4.2 **Tripoli Insurance Compliance**

A team flying under Tripoli Launch Insurance shall comply with DTEG Section 3.

NOTE: Personal, travel, or individual insurance policies do not satisfy the liquid-category requirement. ESRA does not locate insurance policies for teams. Tripoli policy information is available at http://www.tripoli.org/Insurance.

# 5.0 Event Participation and Media Requirements

## 5.1 Event Name, Logos, Links, and Sponsors

R&R-5.1.1 **Official Event Name**

Teams and team sponsors shall refer to the event as the "International Rocket Engineering Competition."

R&R-5.1.2 **Event Logos**

Teams shall use current event logos and applicable style guidance from the ESRA website.

R&R-5.1.3 **ESRA Website Link**

A team-controlled website shall include a link to https://www.esrarocket.org/.

R&R-5.1.4 **Lead Sponsor Recognition**

Teams shall acknowledge the event's Lead Sponsor in their publicity using the current sponsor information maintained by ESRA.

## 5.2 Media and Sponsor Conduct

R&R-5.2.1 **Media Personnel Badging**

Non-student personnel primarily performing media functions shall obtain Media badging as required by the Range Standard Operating Procedures.

R&R-5.2.2 **Team Media Responsibility**

A team shall be responsible for the conduct of media personnel dedicated to covering that team.

R&R-5.2.3 **Protection of Event Reputation**

A team and its sponsors shall not engage in activity that harms or detracts from the standing of the event or event sponsors.

R&R-5.2.4 **Team and Organizer Distinction**

A team and its sponsors shall not imply that they organize or sponsor the event as a whole.

R&R-5.2.5 **Sponsor Conduct Responsibility**

A team shall be responsible for the conduct of its sponsors during the event.

# 6.0 Evaluation, Scoring, and Awards

## 6.1 Category Place Award Eligibility

R&R-6.1.1 **Category Award Eligibility**

A team shall be eligible for a category place award only after a successful launch toward the category target altitude and a successful recovery, unless competition officials exercise the discretionary exception in Section 6.1.

First-place awards are made in each scored category, second-place and third-place awards may be made when sufficient teams compete. When no team satisfies the normal eligibility criterion, competition officials may make category awards based on points accrued, launch attempts, and flight performance and can chose to award no award as well.

## 6.2 SCORING ALLOCATION

**Table 6-1 -** Competition Scoring Allocation

| SCORED ELEMENT | MAXIMUM POINTS | PERCENT OF 1,000 | BASIS |
|---|---|---|---|
| Entry Form and Progress Updates | 60 | 6% | Four pass/fail submissions, 15 points each |
| Project Technical Report | 200 | 20% | Completeness 20; technical communication/format and analysis criteria in Appendix B |
| Design and Implementation | 240 | 24% | Design quality, integration, team knowledge, build quality, verification readiness, and operations |
| Flight Performance | 500 | 50% | Altitude accuracy 350; successful recovery 150 |

## 6.3 ENTRY FORM AND PROGRESS UPDATE SCORING

R&R-6.3.1 **Progress Submission Points**

Each of the Entry Form and three Progress Updates shall be worth 15 points and shall receive those points only when the submission is correct, complete, and timely.

R&R-6.3.2 **Late Entry Form or Progress Update**

A late Entry Form or Progress Update shall disqualify the team from competition.

## 6.4 PROJECT TECHNICAL REPORT SCORING

R&R-6.3.1 **Timely Report Evaluation**

Only a timely Project Technical Report shall be evaluated and scored.

R&R-6.3.2 **Late Report Disqualification**

A late Project Technical Report shall disqualify the team from competition.

R&R-6.3.3 **File Naming Convention**

A Project Technical Report filename shall use the TeamID_Formal University Name_Purpose naming convention; a nonconforming filename shall incur a 5-point deduction.

R&R-6.3.4 **Technical Report Rubric**

The Project Technical Report shall be scored using the Technical Report Rubric in Appendix B.

NOTE: Technical report feedback is not guaranteed. Judges may discuss improvement opportunities during the event but will not disclose final report scores before publication of the Final Score Sheet. A report score may be adjusted when event interactions show that the team's conceptual understanding differs materially from the submitted report.

## 6.5 Design and Implementation Scoring

R&R-6.5.1 **Design and Implementation Evaluation**

Design and Implementation shall be evaluated through the Poster Session, interactions with the team, inspection of the integrated system, and field preparation activities.

R&R-6.5.2 **Design and Implementation Rubric**

Design and Implementation shall be scored using the rubric in Appendix B.

## 6.6 Flight Performance Scoring

R&R-6.6.1 **Flight Performance Scoring**

Flight-performance scoring shall award up to 350 points for altitude accuracy and 150 points for successful recovery.

For an achieved apogee within ±30 percent of the target apogee, altitude points are calculated as:

$$\text{Altitude Score} = 350 \times [1 - |\text{Apogee}_{\text{Target}} - \text{Apogee}_{\text{Actual}}| \div (0.3 \times \text{Apogee}_{\text{Target}})]$$

R&R-6.5.2 **Official Altitude Reporting**

The team shall report the official altitude to Postflight Data Recovery officials immediately after rocket recovery.

R&R-6.5.3 **Telemetry-Based Apogee**

Telemetry-derived apogee may be used for scoring only when it is from the COTS official-altitude sensor source, nominal ascent and recovery events can be confirmed, and the launch vehicle is not recovered before the end of eligible launch operations.

R&R-6.5.4 **Successful Recovery**

Recovery shall be scored successful only when the launch vehicle does not sustain excessive damage as defined in Appendix A.

R&R-6.5.5 **Rapid Fin or Boat-Tail Replacement**

Competition officials may allow replacement of a damaged fin or boat tail designed for rapid replacement when the replacement part is on hand and can reasonably be installed within 30 minutes.

## 6.7 Penalties and Bonuses

R&R-6.7.1 **Conduct Point Deduction**

Each recorded instance of unsafe or unsportsmanlike conduct shall deduct 20 points from the team's earned score.

R&R-6.7.1 **Payload Requirement Penalty**

A violation of the payload requirements for mass, independent function, location and interface, restricted materials, or form factor shall deduct 100 points from the team's earned score.

R&R-6.7.1 **Payload Mass Penalty**

A team shall not add mass to its payload at the event solely to avoid a payload-mass penalty.

R&R-6.7.1 **Payload Form-Factor Bonus**

A functional payload that qualifies for the form-factor exemption and nevertheless conforms to the Payload Cube Unit standard shall receive a 50-point bonus.

Poker chips presented during the competition are on-the-spot recognition and do not add points to the team's score.

## 6.8 Overall and Technical Achievement Awards

The overall New Horizons trophy is awarded to one first-place category winner based on quantitative and qualitative assessments made throughout the event.

**Table 6-2 -** Technical Achievement Awards

| AWARD | PRIMARY BASIS | ELIGIBILITY |
|---|---|---|
| Jim Furfaro Award for Technical Excellence | Overall engineering discipline, analysis, conclusions, planning, execution, procedures, manufacturing, iterative improvement, systems engineering, and robust design. | Accepted into and participates in the Podium Session; deference to a successful launch. |
| Dr. Gil Moore Award for Innovation | A genuinely novel or inventive component, assembly, analytic process, or operational process that solves a unique problem. | Accepted into and participates in the Podium Session; deference to a launch attempt. |
| Charles Hoult Award for Modeling & Simulation | Excellence in mathematical modeling and computational analysis. | Accepted into and participates in the Podium Session; deference to a launch attempt. |
| James Barrowman Award for Flight Dynamics | Smallest percent error between declared predicted apogee and actual apogee. | Successful launch attempt with usable apogee data. |

NOTE: For the first three technical achievement awards, the award decision must be based on material covered in the Podium Session, with the Project Technical Report used as supporting context.

## 6.9 TEAM CONDUCT AND COMMUNICATIONS AWARDS

ESRA may recognize Team Sportsmanship, Nancy Squires Team Spirit, Team Video Challenge, and Live Rocket Video Challenge achievements. Eligibility and selection are based on conduct, event participation, communications quality, technical effectiveness, and current IMS or challenge instructions, as applicable.

# 7.0 Conduct, Disqualification, and Withdrawal

## 7.1 Unsafe and Unsportsmanlike Conduct

R&R-7.1.1 **Range Conduct**

A team and its representatives shall comply with the IREC Range Standard Operating Procedures, use required printed checklists, obey vehicle-safety rules, use required personal protective equipment, and follow reasonable instructions from competition officials.

R&R-7.1.1 **Unsportsmanlike Conduct**

A team representative shall not threaten or show hostility toward an IREC participant or competition official, intentionally misrepresent facts, or intentionally refuse a reasonable instruction from a competition official.

## 7.2 Disqualification

R&R-7.2.1 **Award Consideration Disqualification**

A team may be disqualified from award consideration for failure to meet the defining mission requirements, failure to provide required project details, failure to submit the Project Technical Report or Progress Updates by the required deadlines, or failure to send eligible team representatives to the event.

R&R-7.2.2 **Repeated Conduct Violations**

A team accumulating three conduct violations or 60 total penalty points for safety or unsportsmanlike conduct shall be disqualified.

R&R-7.2.3 **Severe Individual Violation**

An individual committing a single severe safety or unsportsmanlike-conduct violation may be removed and barred from further participation in the event.

## 7.3 Withdrawal and Fees

R&R-7.3.1 **Formal Withdrawal**

A team formally withdrawing before the event shall email general.info@esrarocket.org, comprules@esrarocket.org and launchops@esrarocket.org with a subject identifying the Team ID, formal withdrawal action, and competition year.

R&R-7.3.2 **Non-Refundable Application Fee**

Team application fees shall be non-refundable.

R&R-7.3.3 **Project and Rocketeer Fee Refunds**

Project and Rocketeer fees shall be refundable only until the deadline in the IMS.

R&R-7.3.4 **Late Fee Payment**

A team that pays a required fee late may receive a point deduction or be removed from the competition.

# 8.0 EXPORT-CONTROL REQUIREMENTS

R&R-8.0.1 **ITAR-Free Content**

All IREC submissions, presentations, discussions, and displayed technical content shall be suitable for an ITAR-free event.

R&R-8.0.2 **U.S. Person Responsibility**

A U.S. person participating in IREC shall ensure that submitted or presented content is free of export restrictions and shall not disclose ITAR-controlled information to non-U.S. nationals.

R&R-8.0.3 **Fundamental Research and ITAR**

U.S.-person authors shall ensure Project Technical Reports, Extended Abstracts, and presentation slides remain within their institution's interpretation of fundamental research and applicable ITAR requirements.

NOTE: Additional information is available at https://www.pmddtc.state.gov/ddtc_public.

# 9.0 FINAL SCORE SHEET AND FEEDBACK INQUIRY PROCESS

R&R-9.0.1 **Final Score Sheet Release**

Competition officials shall release the Final Score Sheet no earlier than three weeks after the closing ceremonies.

R&R-9.0.2 **Final Score Sheet Publication**

The Final Score Sheet shall be published in PDF format at https://www.esrarocket.org/2027irec or the current-year equivalent location.

R&R-9.0.3 **Score Inquiry Submission**

A score-sheet inquiry shall be emailed to general.info@esrarocket.org, comprules@esrarocket.org and launchops@esrarocket.org with a subject identifying the Team ID, Final Score Sheet inquiry, and competition year.

R&R-9.0.4 **Score and Feedback Inquiry Deadline**

A team shall submit a Final Score Sheet inquiry within seven days after the score sheet is released.

R&R-9.0.5 **Score Inquiry Conduct**

Score-sheet inquiries and responses shall be professional; rude or unprofessional conduct may incur a 20-point unsportsmanlike-conduct penalty and may invoke Section 7.2.

R&R-9.0.6 **Final Scoring Record**

After the seven-day inquiry period and resolution process, the published scores shall become the final scoring record. This includes any technical or scoring submission feedback.

# 10.0 APPENDIX A — ACRONYMS, ABBREVIATIONS, AND TERMS

## 10.1 ACRONYMS AND ABBREVIATIONS

| ACRONYM | DEFINITION |
|---|---|
| AGL | Above Ground Level |
| AIAA | American Institute of Aeronautics and Astronautics |
| APCP | Ammonium Perchlorate Composite Propellant |
| APRS | Automatic Packet Reporting System |
| CFR | Code of Federal Regulations |
| CONOPS | Concept of Operations |
| COTS | Commercial Off-the-Shelf |
| DTEG | Design, Test, & Evaluation Guide |
| ESRA | Experimental Sounding Rocket Association |
| FAA | Federal Aviation Administration |
| FOR | Flyer of Record |
| GPS | Global Positioning System |
| HPR | High Power Rocket or Rocketry |
| IREC | International Rocket Engineering Competition |
| IMS | Integrated Master Schedule |
| ITAR | International Traffic in Arms Regulations |
| LOX | Liquid Oxygen |
| NAR | National Association of Rocketry |
| RSO | Range Safety Officer |
| SDL | Space Dynamics Laboratory |
| SRAD | Student Researched & Developed |
| STEM | Science, Technology, Engineering, and Mathematics |
| TBD | To Be Determined |
| TBR | To Be Resolved |
| TRA | Tripoli Rocketry Association |
| UAS | Unmanned Aircraft System |

## 10.2 TERM DEFINITIONS

| TERM | DEFINITION |
|---|---|
| Amateur Rocket | An unmanned rocket meeting the limits defined by 14 CFR Part 1, §1.1, including combined total impulse not greater than 889,600 N·s and an altitude capability not greater than 150 km above Earth's surface. |
| Excessive Damage | Damage such that the vehicle could not be safely launched again after replenishing intended consumables. Examples include body-tube zippering greater than one caliber, crushed body tubes, significant avionics-bay, bulkhead, motor-retention, or nose-cone damage. |
| FAA Class 2 Amateur Rocket | An amateur rocket other than a model rocket, propelled by a motor or motors with combined total impulse not greater than 40,960 N·s, as defined by 14 CFR Part 101, §101.22. |
| Intended Consumables | Items reasonably expected to be serviced or replaced following a nominal mission, including propellants, pressurizing gases, and energetic devices. The term may include rapidly replaceable fins or boat tails when replacements are available and can be installed within 30 minutes. |
| Non-toxic Propellant | For IREC purposes, propellants such as APCP, potassium nitrate/sugar, nitrous oxide, LOX, hydrogen peroxide, kerosene, propane, and similar materials that do not require breathing apparatus, extraordinary storage/transport infrastructure, or extensive specialized PPE solely because of toxicity. |
| Payload | A mass or functional item that can be replaced by ballast of the same mass without changing the launch vehicle's ability to reach target apogee or recover successfully. |
| Requirement Verification | The process of showing that the engineered system or team product satisfies a specified requirement. |
| System Validation | The process of showing that the integrated project is suitable for the intended competition mission and stakeholder use. |

# 11.0 APPENDIX B — JUDGING RUBRICS

Judging Principle: The rubrics summarize the content required by this specification. Judges should score the quality, completeness, and technical credibility of the submitted work without introducing criteria that are not requested in the applicable report, presentation, or design sections.

The rubrics in this appendix are aligned to the body requirements. Judges should score the evidence required by the specification rather than introduce criteria that are not requested from teams. The “Assumptions and Sensitivity Analysis” criterion in the baseline has been replaced by an explicit body requirement and a rubric criterion titled “Assumptions, Uncertainty, Margins, and Sensitivity.”

## 11.1 Project Technical Report Rubric — 200 Points

**Table 11-1 -** Project Technical Report Rubric

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Completeness and Required Content (20)** | All required sections, appendices, data, figures, and supporting material are present and easy to locate. Non-applicable items are clearly identified. | Minor omissions do not prevent a complete evaluation. | Several required items are incomplete, unclear, or difficult to locate. | One or more major required sections or appendices are missing. | /20 |
| **Technical Communication, Mechanics, and Format (25)** | Clear, concise, well organized, technically edited, consistent units and terminology, effective figures and tables, current template used, and page limits met. | Generally clear and professional with only minor errors or template deviations. | Readable, but recurring organization, grammar, unit, figure, or format problems reduce effectiveness. | Unclear, poorly organized, technically unreliable, or substantially noncompliant with format or page limits. | /25 |
| **Mission, Objectives, Team Organization, and CONOPS (25)** | The competition category, mission, objectives, payload purpose, team organization, management approach, mission phases, and phase transitions form a clear and consistent project story. | Most elements are clear and consistent; minor gaps remain. | The project story is understandable, but several elements are incomplete or weakly connected. | The mission, objectives, organization, or operations are unclear or contradictory. | /25 |
| **System Architecture, Interfaces, and Design Rationale (35)** | The integrated vehicle and all applicable subsystems are clearly defined. Important interfaces, design choices, alternatives, loads, constraints, and safety factors are explained with sound engineering rationale. | Most major design decisions and interfaces are well supported; minor gaps remain. | The design is described, but important interfaces, alternatives, or engineering rationale are weak. | The report is primarily descriptive and key design decisions are unsupported or inconsistent. | /35 |

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Analysis, Assumptions, Margins, and Sensitivity (30)** | Analyses are technically sound; important assumptions and uncertainties are stated; design margins are quantified; sensitivity analysis identifies the variables that most affect the result and supports design decisions. | Analyses are generally sound with small gaps in assumptions, uncertainty, margins, or sensitivity. | Analysis supports some decisions, but assumptions or uncertainty are often unstated and margin or sensitivity treatment is limited. | Analysis is sparse, incorrect, contradictory, or disconnected from design decisions. | /30 |
| **Testing and Evidence (30)** | Test purpose, configuration, acceptance criteria, results, anomalies, conclusions, and resulting design changes are complete and credible. Evidence supports the major design and readiness claims. | Most major claims have appropriate evidence; minor gaps remain. | Some useful evidence is provided, but test purpose, criteria, results, or conclusions are incomplete. | Testing and supporting evidence are inadequate, unclear, or technically unreliable. | /30 |
| **Safety, Risk, Operations, and Lessons Learned (35)** | Hazards, mission risks, mitigations, checklists, off-nominal procedures, reliability considerations, lessons learned, and knowledge transfer are complete and connected to the design and CONOPS. | Coverage is generally complete with minor gaps. | Coverage is partial or only loosely connected to the design and operations. | Major safety, risk, checklist, or lessons-learned content is missing or not credible. | /35 |
| **Total** | | | | | **/200** |

## 11.2 Design and Build Rubric — 240 Points

**Table 11-2 -** Design and Build Rubric

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Mission Fit and Integrated Design (30)** | The team presents a coherent, achievable mission and the completed hardware and operations clearly support the stated category, objectives, and competition requirements. | The design generally supports the mission with minor gaps. | The mission is only partly reflected in the implemented design. | The design conflicts with major mission goals or competition requirements. | /30 |
| **Subsystem and Interface Integration (30)** | Subsystems and interfaces are well integrated, accessible, compatible, and ready for operation; late changes are controlled and understood. | Integration is generally sound with minor interface issues. | Several interfaces or late changes are weakly integrated. | Major incompatibilities or uncontrolled field modifications threaten mission success. | /30 |
| **Engineering Rationale and Team Knowledge (35)** | Team members demonstrate broad understanding of the physical principles, analyses, tradeoffs, margins, and reasons behind the design. | Good understanding with only limited reliance on a few specialists. | Understanding is uneven and the team frequently defers to one or two experts. | The team cannot explain governing principles or key design decisions. | /35 |
| **Robustness, Margins, and Mission Suitability (35)** | The design is robust for expected launch, flight, recovery, and handling conditions; margins and failure tolerance are demonstrated. | The design is suitable with modest robustness or margin gaps. | The design may work under narrow conditions but is sensitive to expected variation. | The design is unlikely to operate safely or successfully under expected conditions. | /35 |
| **Manufacturing, Workmanship, and Configuration (30)** | Construction methods are appropriate and correctly applied; workmanship is excellent; the inspected hardware matches the documented design. | Good workmanship and methods with minor quality or documentation issues. | Methods are generally appropriate but inconsistently applied or documented. | Methods, workmanship, or hardware configuration are inadequate or impractical. | /30 |

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Test Evidence and Flight Readiness (30)** | Inspection, analysis, test, and demonstration evidence support readiness; anomalies are resolved; the flight hardware matches the tested configuration. | Evidence supports most critical functions with minor unresolved issues. | Evidence is incomplete or its connection to the current hardware is weak. | Critical functions lack credible evidence or unresolved problems prevent confidence. | /30 |
| **Safety, Checklists, and Operations (30)** | The team operates safely and efficiently; required checklists are complete, available, understood, and used correctly; off-nominal actions are practical. | Generally safe and disciplined with minor issues. | Minimum compliance with recurring preparedness or procedural weaknesses. | Unsafe or noncompliant practices; required checklists or procedures are absent or unused. | /30 |
| **Operational Execution and Professionalism (20)** | The team communicates clearly, prepares efficiently, responds professionally, and demonstrates ownership throughout judging and field activities. | Generally effective and professional with minor lapses. | Preparation, communication, or ownership is inconsistent. | The team is unprepared, unprofessional, or unable to execute normal operations. | /20 |
| **Total** | | | | | **/240** |

## 11.3 Podium Session Rubric — 100 Points

The Project Technical Report may be consulted for context; however, technical achievement awards shall be based on the material actually presented and defended during the Podium Session.

**Table 11-3 -** Podium Session Scoring and Award-Consideration Rubric

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Technical Significance and Problem Definition (15)** | An important technical problem is clearly defined and tied to the launch-vehicle mission or a meaningful engineering need. | A useful problem with clear mission relevance. | The problem is somewhat relevant but incompletely framed. | The problem is unclear, trivial, or not launch-vehicle related. | /15 |
| **Technical Depth, Evidence, and Conclusions (25)** | Analysis or test evidence is rigorous; assumptions, methods, results, limitations, and conclusions are credible and connected. | Strong evidence with minor gaps. | Some useful evidence is presented, but technical depth or conclusions are incomplete. | Assertions are unsupported or technically unsound. | /25 |
| **Engineering Decision Process (15)** | Alternatives, interfaces, risks, assumptions, and evidence are integrated into a clear explanation of why the team selected its approach. | Most decision logic is coherent. | Some decision logic is present, but important connections are weak. | Little evidence of a structured engineering decision process. | /15 |
| **Innovation or Novel Contribution (15)** | A genuinely novel, inventive, or unusually effective solution addresses a clearly defined problem. | A meaningful improvement or creative application is demonstrated. | The contribution is incremental with limited novelty. | No identifiable innovation or technical contribution is demonstrated. | /15 |
| **Modeling and Simulation Quality (15)** | Models are appropriate, explained, checked against data where practical, and used insightfully to predict or explain behavior. | Good modeling with minor limitations. | Basic modeling with limited validation or insight. | Modeling is absent, inappropriate, or misleading. | /15 |

| CRITERIA | OUTSTANDING | EXCELLENT | SATISFACTORY | UNSATISFACTORY | SCORE |
|---|---|---|---|---|---|
| **Presentation, Visuals, and Q&A (15)** | The presentation is professional, clear, on time, visually effective, and supported by authoritative answers to questions. | Clear presentation and good responses with minor weaknesses. | Understandable but uneven delivery, visuals, timing, or Q&A. | Difficult to follow; poor visuals or timing; key questions cannot be answered. | /15 |
| **Total** | | | | | **/100** |

**Table 11-4 -** Podium Award Consideration

| AWARD | JUDGE QUESTION | ELIGIBLE? | SPECIFIC EVIDENCE / HIGHLIGHT |
|---|---|---|---|
| Furfaro Technical Excellence | Did the presentation demonstrate exceptional overall engineering discipline, execution, systems engineering, robust design, and technical skill? | Yes / No | |
| Moore Innovation | Did the presentation demonstrate a genuinely novel, inventive, or uniquely problem-solving feature or process? | Yes / No | |
| Hoult Modeling & Simulation | Did the presentation demonstrate excellence in mathematical modeling and computational analysis? | Yes / No | |

# 12.0 APPENDIX C — TEAM VIOLATION POINT DEDUCTION FORM

Use one form for each recorded violation. Each selected violation deducts 20 points. Three recorded instances or 60 accumulated violation points result in team disqualification.

## 2027 IREC TEAM VIOLATIONS POINT FORM

| Date: 06 / /2027 | Time: _____ | Team #: _____ |
|---|---|---|

**VIOLATIONS**(-20 PTS. For each box selected, If Team is found with 3 instances of 60 in violations team is disqualified)

**Unsafe Conduct:**
- ☐ Violating IREC Range Standard Operating Procedures.
- ☐ Losing student badge and not having required TRA info to get a new badge.
- ☐ Failure to use printed checklists during operations.
- ☐ Violating ESRA / local motor vehicle traffic safety rules at launch site.
- ☐ Failure to use appropriate personal protective equipment.
- ☐ Failure to wait until ESRA Recovery releases team to recover.
- ☐ Failure to return directly to ESRA Recovery tent to check in and return equipment.
- ☐ Other:_____

**Unsportsmanlike Conduct:**
- ☐ Hostility/ threats shown towards any IREC Participant or competition official.
- ☐ Intentional misrepresentation of facts to any competition official.
- ☐ Intentional failure to comply with any reasonable instruction given by a competition official.
- ☐ Switching Teams Assigned Conference Table without ESRA permission
- ☐ Team not attending Conference Day
- ☐ Other: _____

**Unpreparedness**
- ☐ Taking longer than 30 minutes to set up rocket on rail to go vertical, once arrived at the pads.
- ☐ Taking longer than 45 minutes in Post Flight Data Recovery with Judge failure to bring tools and needed equipment even after warnings.

| **Competition Official Observing Violation:** ☐ ESRA Judge ☐ ESRA Range Safety ☐ ESRA Postflight ☐ Awards ☐ SDL ☐ ESRA Registration ☐ ESRA Livestream/Media ☐ ESRA Leadership ☐ ESRA Recovery | **Print Name:** _____ |
|---|---|
| **Initials of Signoff In score sheet & Date:** | **Total pts. Deducted:** |

**Provide any comments/notes on back of sheet**